

RETRIVAL PIPELINE

CREATE A NEW VECTOR INDEX, THIS TIME ITS FOR THE BM25, SO USE THE ATLAS SEARCH OPTION

Start Your Index Configuration

Search Type

Which Search type should I use?

Atlas Search

Full-text search for relevance-based app features.

Vector Search

For semantic search and AI applications.

Index Name and Data Source

Search indexes are specific to a database and collection.

At this time, search indexes cannot be created for time series collections.

Index Name

bm25_search

Database and Collection

Search for database or collection ...

db_testcases

test_cases

Configuration Method

Select how you would like to configure your index. You can also define indexes through the [Atlas Admin API](#), [Atlas CLI](#), [MongoDB Shell](#), or supported [Drivers](#).

Visual Editor

Define index definitions in a more guided experience.

JSON Editor

Edit the raw index definition with an embedded JSON editor.

UPDATE INDEX.JSON VALUE

```

49     "linkedStories": {
50         "type": "string",
51         "analyzer": "lucene.keyword"
52     },
53     "createdBy": {
54         "type": "string",
55         "analyzer": "lucene.keyword"
56     },
57     "version": {
58         "type": "string",
59         "analyzer": "lucene.keyword"
60     }
61 }
62 }
63 }

```

```

6     "type": "string",
7     "analyzer": "lucene.keyword"
8 },
9 "module": {
10     "type": "string",
11     "analyzer": "lucene.standard"
12 },
13 "title": {
14     "type": "string",
15     "analyzer": "lucene.standard"
16 },
17 "description": {
18     "type": "string",

```

[Back](#)
[Cancel](#)
[Create Search Index](#)

Index Name	Status	Fields
db_testcases		
test_cases		
bm25_search	PENDING	"automationManuc", "createdBy", "description", "expectedResults", "id" + 9 more
	View status details	

THE INDEX WOULD BECOME ACTIVE IN FEW SECONDS

Index Name	Status	Fields
db_testcases		
test_cases		
bm25_search	READY	"c", "c", "c", "e", "i"
	View status details	

THIS INDEX ALREADY REFERS TO THE TEST_CASES COLLECTION TO WHICH WE HAVE ALREADY INJECTED WITH DATA

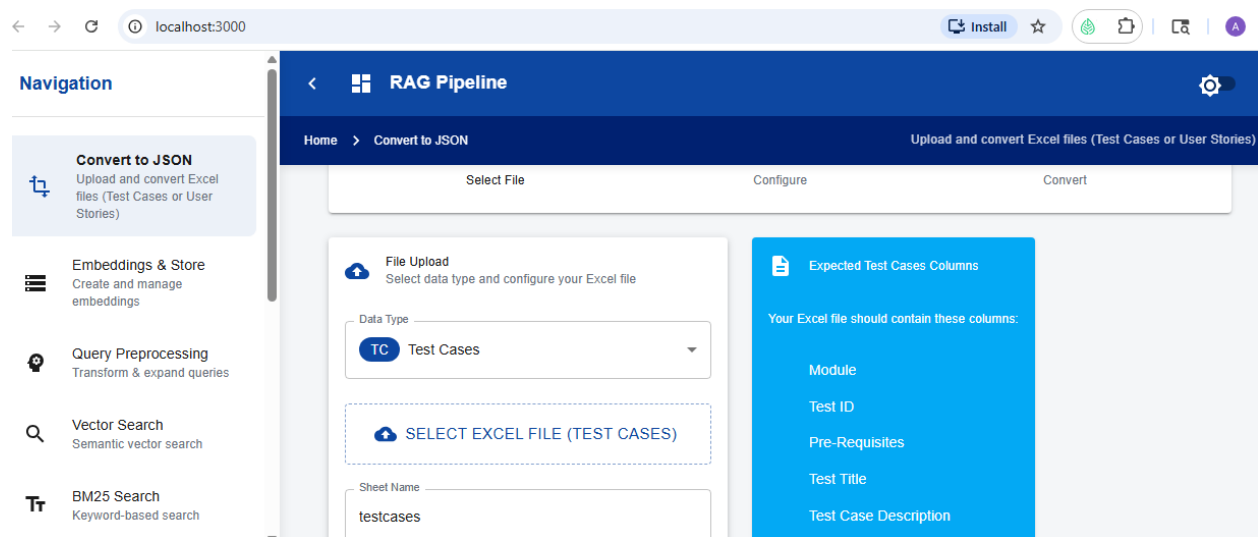
NOW WE HAVE TO UPDATE THE ENV FILE IN OUR APPLICATION

```

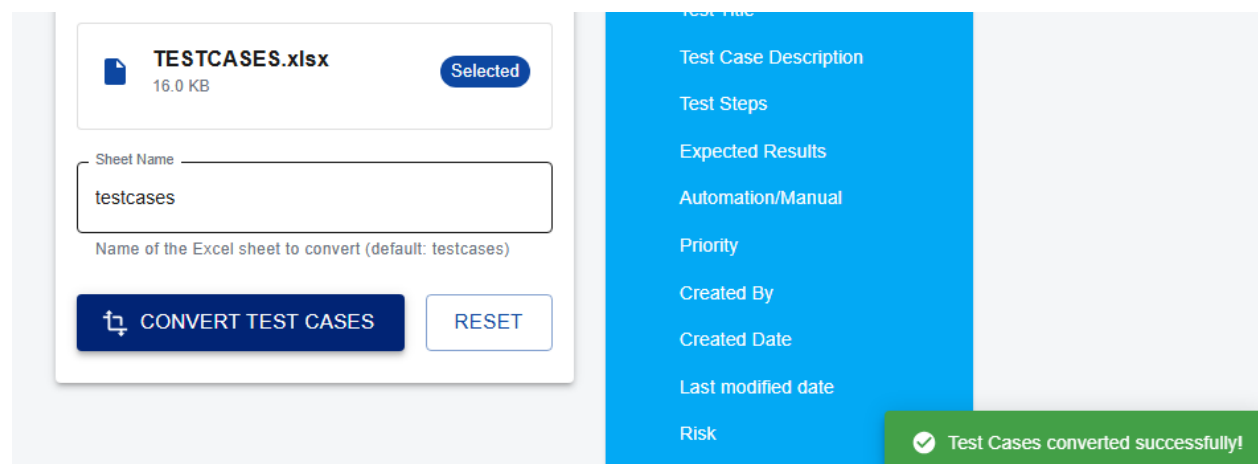
1 MONGODB_URI=mongodb+srv://iamanandhiram_db_user:iamanandhiram@ananindia.ohf5xrt.mongodb.net/?appl
2 #MONGODB_URI=mongodb+srv://sumitha:TestLeaf321@cluster0.c5L4wjq.mongodb.net/?retryWrites=true&w=
3 DB_NAME="db_testcases"
4 #DB_NAME="RagTest_cases"
5
6 COLLECTION_NAME="test_cases"
7 VECTOR_INDEX_NAME="vector_index_test_cases"
8 BM25_INDEX_NAME="bm25_search"
9
10 USER_STORIES_COLLECTION_NAME="user_stories"
11 USER_STORIES_VECTOR_INDEX_NAME="vector_index_user_story"
12 USER_STORIES_BM25_INDEX_NAME="bm25_user_stories"
13
14 #USER_STORIES_COLLECTION_NAME="user_stories"
15 #USER_STORIES_VECTOR_INDEX_NAME="vector_index"

```

ONCE DONE LETS LOCALLY START THE APPLICATION USING NPM INSTALL AND NOM RUN DEV COMMANDS. THE APPLICATION IS LAUNCHED NOW



LETS UPLOAD THE TEST CASE FILE WE HAVE AND CREATE EMBEDDING



✓ Success! Test cases file converted successfully

Output File

converted-1779693925657.json

Saved to data folder

Status

Completed

Ready for embedding creation

Conversion Log

⚠ Sheet "testcases" not found. Using first sheet: "Sheet1"
✓ Converted 66 rows from "Sheet1" into D:/Users/a.natarajan/OneDrive - Corpay/personal/rag-mongo-demo-v9/rag-mongo-demo-v9/src/data/converted-1779693925657.json

CONVERT THE JSON TESTCASE FILE TO EMBEDDINGS AND PUSH TO THE VECTOR DB

Embeddings & Store
Create and manage embeddings

Query Preprocessing
Transform & expand queries

	File Name	Size	Modified ↓	Type
✓	converted-1779693925657.json	30.55 KB	5/25/2026 12:55:26 PM	JSON

Current file: testcases

Progress: 0%

Job ID: job-1779696017575-d4...

🔔 This process continues even if you refresh the page!

Actions
Process selected files

PROCESSING...

✓ Batch processing started! Job ID: job-1779696017575-d48xoss64

🔔 Starting batch processing for 1 files...

Processing Results

1 completed, 0 failed



testcases completed

Input file: src/data/converted-1779693925657.json
MISTRAL AI BATCH PROCESSING: 66 test cases
Using Mistral AI Embedding API for Cost-Effective Processing!
Configuration:
Batch Size: ...

LETS TRY TO CHECK THE COLLECTION FOR THE TEST CASES

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

+

25 1 - 25 of 50

```
_id: ObjectId('6a1403ad304ddce46b9fb4d4')
module: "UPI Payments"
id: "TC_001"
```

PERFORM THE SAME FOR USERSTORIES AS WELL

 **USERSTORIES.xlsx** Selected

8.6 KB

Sheet Name

stories

Name of the Excel sheet to convert (default: stories)

 CONVERT USER STORIES

RESET

Output File

[stories-1779696673635.json](#)

Saved to data folder

Status

Completed

Ready for embedding creation

Conversion Log

⚠ Sheet "stories" not found. Using first sheet: "Sheet1"
🔄 Converting 5 rows from sheet "Sheet1"...
✅ Converted 5 user stories from "Sheet1" into D:/Users/a.natarajan/OneDrive - Corpay/personal/rag-mongo-demo-v9/rag-mongo-demo-v9/src/data/stories-1779696673635.json
⚠ Filtered out: 0 empty rows

LETS CONVERT THIS JSON TO EMBEDDINGS AND PUSH TO VECTOR DB COLLECTION USERSTORY

Output File

[stories-1779696673635.json](#)

Saved to data folder

Status

Completed

Ready for embedding creation

Conversion Log

⚠ Sheet "stories" not found. Using first sheet: "Sheet1"
🔄 Converting 5 rows from sheet "Sheet1"...
✅ Converted 5 user stories from "Sheet1" into D:/Users/a.natarajan/OneDrive - Corpay/personal/rag-mongo-demo-v9/rag-mongo-demo-v9/src/data/stories-1779696673635.json
⚠ Filtered out: 0 empty rows

Available Files

11 JSON files found



REFRESH

	File Name	Size	Modified ↓	Type
☒	stories-1779696673635.json	4.04 KB	5/25/2026 1:41:14 PM	JSON

Processing Embeddings...0/1

Current file: user-stories

Progress: 0%Job ID: job-1779696764319-cj...

This process continues even if you refresh the page!

Actions

Process selected files

✓ Batch processing started! Job ID: job-1779696764319-cj2k53xjw

i

Starting batch processing for 1 files...

Processing Results

1 completed, 0 failed

✓ testcases completed

📁 Input file: src/data/converted-1779697697433.json

🚀 MISTRAL AI BATCH PROCESSING: 5 test cases

⚡ Using Mistral AI Embedding API for Cost-Effective Processing!

⚙ Configuration:

📦 Batch Size: 5...

DATA IS INGESTED INTO THE VECTOR DB UNDER THE SPECIFIED COLLECTION NAME

CLUSTERS (1)

Search clusters

ANANINDIA

admin

db_testcases

test_cases

user_stories

local

Type a query: { field: 'value' } or [Generate query](#)

ExplainResetFindOptions

251 - 5 of 5

_id: ObjectId('6a1410f105ac57394eb3faf7')

key: "US_001"

summary: "UPI Payment to valid ID"

description: "As a user, I want to send money to a valid UPI ID so that I can comple..."

status: Object

priority: Object

assignee: null

ADDED 2 ABBREVIATIONS TO THE ABBREVIATIONMAPPING.JS FILE AND DICTIONARIES.JS FILE

```
//TRANSACTION PROCESSING
🔗"UPI" : "unified payments interface",
"PIN" : "personal identification number",
"₹" : "rupees"
```

```
7  export const abbreviationMap = {
75
76    // Insurance
77    "ins": "insurance",
78    "preauth": "pre authorization",
79
80    // Pharmacy
81    "pharm": "pharmacy",
82    "med": "medication",
83    "drug": "medication",
84
85    //TRANSACTION PROCESSING
86    "UPI" : "unified payments interface",
87    "PIN" : "personal identification number"
88  };
89
90  // Healthcare Domain Synonyms
91  export const synonymMap = {
92    // Person/Role
```

LET'S PERFORM QUERY PROCESSING THAT WOULD INCLUDE THE ABBREVIATIONS WE HAVE CONFIGURED

Query Preprocessing & Search

Transform your query through normalization, abbreviation expansion, and synonym generation before searching

Enter Your Query

OTP based authentication before processing a UPI transaction

Quick Examples:

UHID patient login issue OTP not working

TC_027 merge UHID records

doctor appointment booking IP admission

password reset OTP verification

ER patient registration BP monitoring

 PREPROCESS QUERY

 ORIGINAL QUERY

"OTP based authentication before processing a UPI transaction"

1 NORMALIZATION

"otp based authentication before processing a upi transaction"

8 tokens

2 ABBREVIATION EXPANSION **2 expansions**

"one time password based authentication before processing a unified payments interface transaction"

Expanded:

otp → one time password

UPI → unified payments interface

LET'S PERFORM VECTOR SEARCH ON USER STORY AND SEE THE RETRIEVAL BASED ON SEMANTICS

Enter your search query

FAILED UPI PAYMENTS

Use descriptive terms related to test functionality

Advanced Metadata Filters

☐ SHOW FILTERS

Results Limit

5 Results

 SEARCH



Search Query: "FAILED UPI PAYMENTS"

5 results

Performance: \$0.000001 cost 10 tokens

Test Case	Similarity	Description	Source
Failed transaction visibility	91.1%	Check failed list	UPI Payments
Refund on failed transaction	90.4%	Trigger failure	UPI Payments
Invalid UPI ID failure	89.6%	Enter wrong ID	UPI Payments
Verify successful transfer to	89.4%	Enter valid UPI ID and amount	UPI Payments

Detailed View

Failed transaction visibility	 Score: 0.9108	UPI Payments ▾
Refund on failed transaction	 Score: 0.9045	UPI Payments ▾
Invalid UPI ID failure	 Score: 0.8960	UPI Payments ▾
Verify successful transfer to valid UPI ID	 Score: 0.8939	UPI Payments ▾
Network failure during transaction	 Score: 0.8920	UPI Payments ▾

LETS NOW TRY THE SAME QUERY USING BM SEARCH

BM25 Keyword Search

Fast keyword-based search using BM25 algorithm. Best for exact matches, IDs, and specific terms.

Search Query

FAILED UPI PAYMENTS

Results Limit

5

 SEARCH

 SHOW FILTERS ▾

 BM25 Search Results: Found 5 test cases in 14ms

 Found 5 results in 14ms

#1 **TC_047** **Score: 4.71**

Failed transaction visibility

Module: UPI Payments

Medium

Risk: Medium

Manual

Description: Check failed list

Steps: 1. Filter failed

Expected Results: Displayed

 Created: 5/25/2026

#2 **TC_001** **Score: 3.50**

Verify successful transfer to valid UPI ID

Module: UPI Payments

High

Risk: High

Manual

Description: Enter valid UPI ID and amount

#3 **TC_007** **Score: 2.66**

Verify multiple sequential payments

Module: UPI Payments Medium Risk: Medium Manual

Description: Make 3 payments quickly

Steps: 1. Repeat payment thrice

Expected Results: All processed without error

Created: 5/25/2026

#4 **TC_005** **Score: 2.51**

Verify small amount payment

Module: UPI Payments Medium Risk: Medium Manual

Description: Test ₹1 payment

Steps: 1. Enter ₹1 2. Confirm

Expected Results: Payment successful

#5 **TC_028** **Score: 2.29**

Concurrent balance usage

Module: UPI Payments Medium Risk: Medium Manual

Description: Parallel payments

Steps: 1. Initiate 2 payments

Expected Results: Only valid processed

Created: 5/25/2026

HYBRID SEARCH



Hybrid Search (BM25 + Vector)

Combines BM25 keyword matching with semantic vector search for best results. Adjust weights to control the balance.

Search Query

FAILED UPI PAYMENTS

Results Limit

5

SEARCH

Search Weights ⓘ

BM25 Weight: 50%

Vector Weight: 50%



Search Results (5)

#1 **TC_047** Hybrid: 95.5% Found in: both

Failed transaction visibility

Module: UPI Payments Medium Risk: Medium Manual

Description: Check failed list

Steps: 1. Filter failed

Expected Results: Displayed

Score Breakdown:

BM25: 100.0% (raw: 4.71) Vector: 91.1% (raw: 0.91) Hybrid: 95.5%

Created: 5/25/2026

#2 **TC_001** Hybrid: 81.9% Found in: both

Verify successful transfer to valid UPI ID

Module: UPI Payments High Risk: High Manual

Description: Enter valid UPI ID and amount

Steps: 1. Enter valid UPI ID 2. Enter ₹500 3. Enter PIN

Expected Results: Payment successful with confirmation

Score Breakdown:

BM25: 74.3% (raw: 3.50) Vector: 89.4% (raw: 0.89) Hybrid: 81.9%

Created: 5/25/2026

LETS SUMMARIZE THE RESULT AND AVOID ANY DUPLICATE IN THE SEARCH

Search Summarization & Deduplication

Search, remove duplicates, and generate AI-powered summaries of test cases

Search Query

FAILED UPI PAYMENTS

Search Type

Vector

Limit

10

Quick Examples:

patient login authentication

merge UHID records

appointment booking

password reset OTP

SEARCH

DEDUPLICATE

SUMMARIZE WITH AI

COPY ALL RESULTS

Dedup Threshold

0.85

Summary Type

Concise

TC_047 Score: 0.9108 Failed transaction visibility Check failed list UPI Payments Medium Priority Functional	TC_015 Score: 0.9045 Refund on failed transaction Trigger failure UPI Payments Medium Priority Functional
TC_011 Score: 0.8960 Invalid UPI ID failure Enter wrong ID UPI Payments High Priority Functional	TC_001 Score: 0.8939 Verify successful transfer to valid UPI ID Enter valid UPI ID and amount UPI Payments High Priority Functional
TC_017 Score: 0.8920 Network failure during transaction Disconnect internet	TC_016 Score: 0.8900 Incorrect account mapping Wrong details
TC_018 Score: 0.8883 Backend validation error Simulate failure UPI Payments Medium Priority Functional	TC_019 Score: 0.8881 SMS notification failure Block SMS UPI Payments Medium Priority Functional
TC_033 Score: 0.8879 PIN retry attempts 3 wrong attempts UPI Payments High Priority Functional	TC_021 Score: 0.8878 Insufficient balance check Low balance UPI Payments High Priority Functional

I DINT HAVE ANY DUPLICATES

10

SEARCH RESULTS

DEDUPLICATED

AI SUMMARY

Deduplication Statistics

Original Count

After Deduplication

Duplicates Removed

Reduction

10

10

0

0.0%

Unique Results (10)

COPY ALL UNIQUE

TC_047

✓ Unique

Score: 0.9108

Failed transaction visibility

TC_015

✓ Unique

Score: 0.9045

Refund on failed transaction

AI-Generated Summary

Overview of "test cases" Search Results (UPI Payments Module)

| ID | Title (focus) | Core Action | Relevance |

| ---|-----|-----|-----|

| **TC_047** | Failed transaction visibility | Verify that failed transactions appear in the failed-list view | 91.1 % |

| **TC_015** | Refund on failed transaction | Initiate a refund after a transaction failure | 90.4 % |

| **TC_011** | Invalid UPI ID failure | Enter an incorrect UPI ID and expect error handling | 89.6 % |

| **TC_001** | Successful transfer to valid UPI ID | Perform a normal transfer with a correct ID and amount | 89.4 % |

| **TC_017** | Network failure during transaction | Simulate loss of internet connectivity mid-transaction | 89.2 % |

| **TC_016** | Incorrect account mapping | Supply mismatched account details and verify mapping error | 89.0 % |

| **TC_018** | Backend validation error | Force a server-side validation failure | 88.8 % |

| **TC_019** | SMS notification failure | Block SMS delivery and confirm no notification is sent | 88.8 % |

| **TC_033** | PIN retry attempts | Enter a wrong PIN three times to test lock-out behavior | 88.8 % |

| **TC_021** | Insufficient balance check | Attempt a transfer with a low-balance account | 88.8 % |

Key Themes

1. **Error-handling & Visibility** – Several cases (TC_047, TC_015, TC_011, TC_018) focus on how the system reports and processes failures.

2. **Network & External Dependencies** – TC_017 and TC_019 test resilience to connectivity and SMS service disruptions.

3. **Security & Validation** – TC_033 validates PIN-retry limits; TC_016 checks account-mapping integrity.

4. **Positive Flow Baseline** – TC_001 provides a control case for a successful transfer.

5. **Business Rules** – TC_021 ensures the platform correctly blocks transactions when funds are insufficient.

LET'S MODIFY THE SCHEMA AND PROMPT TEMPLATE AND TRY TO GENERATE NEW TEST CASES FOR OUR NEW USER STORY

User Story ID: US_007

Title: Real-time FX Rate Display for Cross-Border UPI Payment

Actor: UPI User

Description:

As a UPI user, I want to view the real-time foreign exchange rate and converted amount before confirming a cross-border payment so that I know exactly how much will be debited in INR and credited in foreign currency.

Preconditions:

- User has a valid UPI account enabled for international payments.
- Destination country and currency are supported.
- FX rate service is available.

Acceptance Criteria:

- System displays real-time FX rate before payment confirmation.
- Converted amount (INR to foreign currency) is clearly shown.
- Any fees or charges are transparently listed.
- User must confirm after viewing FX details.
- Payment proceeds only after user consent to the displayed rate.

JSON Schema Configuration



```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "UPI Transaction Test Case",
  "type": "object",
  "properties": {
    "testCaseId": {
      "type": "string",
      "description": "Unique identifier for the test case, e.g., TC_001"
    },
    "module": {
```

JSON SCHEMA

PROMPT TEMPLATE

TEST & PREVIEW

METRICS EVALUATION

Prompt Template

UPI TRANSACTION TEST CASE GENERATION

INSTRUCTION
Generate 10 high-quality test cases for the user story using retrieved test case context. Each must:
- Include 5-8 detailed, numbered test steps
- Define measurable expected results
- Cover positive, negative, and edge cases
- Reference source test cases

CONTEXT
MongoDB database with 25+ UPI TRANSACTION test cases covering UPI RELATED TRANSACTIONS

Test Prompt Template

Test your prompt template with sample data to see how the AI responds

Test Query / User Story

User Story ID: US_007

Title: Real-time FX Rate Display for Cross-Border UPI Payment

Sample Test Cases (JSON)

[
 {
 "module": "UPI Payments",
 "id": "TC_001",
 "preRequisites": "User logged in & UPI configured",
 "title": "Verify successful transfer to valid UPI ID",
 "description": "Enter valid UPI ID and amount",
 "steps": "1. Enter valid UPI ID\n2. Enter ₹500\n3. Enter PIN",
 "expectedResults": "Payment successful with confirmation",

🧠 Complete RAG Pipeline Results (10-Step Process)

1. User Input → 2. Preprocessing → 3. Hybrid Search → 3.5. User Story Search → 4. RRF Rerank → 5. Dedup → 6. Summarize → 7. Prompt →

50 hybrid search results

RRF Re-Ranking Applied

0 duplicates removed

Top 10 results

✓ Validation Passed

📄 10-Step Pipeline Completed

✓ User Story Input

✓ Query Preprocessing (Normalize → Abbreviations → Synonyms)

✓ Hybrid Search – Test Cases (BM25 + Vect

✓ Hybrid Search – User Stories (BM25 + Vector)

✓ RRF Re-Ranking (Cross-encoder, top 10 selected)

✓ Deduplication (Cosine > 0.

✓ Prompt Template + Context (ICEPOT framework)

✓ LLM Generation (TestLeaf API)

✓ JSON Validation (AJV)

✓ HTML Conversi

TC_001	Verify successful transfer to valid UPI ID	UPI Payments	High	N/A	1. Enter valid UPI ID 2. Enter ₹500 3. Enter PIN
TC_005	Verify small amount payment	UPI Payments	Medium	N/A	1. Enter ₹1 2. Confirm
TC_002	Verify QR code payment	UPI Payments	High	N/A	1. Scan QR 2. Enter amount 3. Confirm
TC_022	Exact balance payment	UPI Payments	High	N/A	1. Enter exact balance
TC_007	Verify multiple sequential payments	UPI Payments	Medium	N/A	1. Repeat payment thrice
TC_028	Concurrent balance usage	UPI Payments	Medium	N/A	1. Initiate 2 payments

Newly generated test cases

Analysis & Gaps

User Story: Real-time FX Rate Display for Cross-Border UPI Payment
Module: UPI Payments – International
Coverage: 10 existing test cases

Gaps Identified:

- Negative-flow validation (invalid currency, insufficient funds, service outage)
- FX-rate volatility handling between fetch and confirmation
- OTP/2FA failure scenarios
- Boundary-value limits (minimum, maximum transaction amount)
- Concurrent/rapid-fire cross-border payments
- UI/UX compliance – formatting, accessibility, fee disclosure

TC_0051	Happy-Path Cross-Border Payment with Real-Time FX Rate	UPI Payments – International	P1	Functional	User has a UPI account with international-payment enablement, sufficient INR balance, destination country (USA) and currency (USD) are supported, FX-rate service is healthy.	1. Enter valid UPI ID and amount (₹5,000). 2. Tap "Get Rate" – system fetches real-time FX rate. 3. Verify that the screen shows: FX rate (e.g., 1 USD = 82.45 INR), converted amount (USD 60.64), and itemised fees. 4. Tap "Get Rate" – system fetches real-time FX rate. 5. Verify that the screen shows: FX rate (e.g., 1 USD = 82.45 INR), converted amount (USD 60.64), and itemised fees. 6. Press "Confirm". 7. Enter UPI PIN.
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TC_0052	FX-Rate Service Unavailable – Graceful Degradation	UPI Payments – International	P1	Integration	FX-rate micro-service is deliberately stopped (simulated via fault injection). User has sufficient balance.	amount (₹2,000). 4. Tap "Get Rate". 5. Observe system response after 5 seconds timeout. 6. Verify error message displayed. 7. Attempt to proceed to confirmation. 8. Capture screenshot of UI.
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Affected / Similar User Stories

These user stories are semantically or functionally related to the current query and may be impacted by changes. Review for regression risks, shared dependencies, or over

0 High Regression Risk

0 Medium Risk

5 Low / Informational

5 Total Stories

US_001 Low Regression Risk

UPI Payment to valid ID

Module: UPI Transactions

Priority: High

Status: To Do

Impact Reason: Potentially impacted via shared functionality | Same epic/feature area: UPI Transactions

Acceptance Criteria: Enter valid UPI ID, amount and PIN; payment succeeds.

US_004 Low Regression Risk

Secure UPI PIN Authentication

Module: UPI Transactions

Priority: High

Status: To Do

Impact Reason: Potentially impacted via shared functionality | Same epic/feature area: UPI Transactions

Acceptance Criteria: Correct PIN required; lock after invalid attempts.

UNABLE TO DO THE AFFECTED USERSTORIES DUE TO LIMITATION OF HAVING JUST 3 INDEX

